06 - Operational Complexity

Due Oct 13 at 2pm **Points** 1 **Questions** 2
Available until Oct 20 at 11:59pm **Time Limit** 30 Minutes **Allowed Attempts** 10

Take the Quiz Again

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	5 minutes	0.85 out of 1

ⓘ Correct answers are hidden.

Score for this attempt: **0.85** out of 1
Submitted Oct 19 at 1:38pm
This attempt took 5 minutes.

Question 1

0.5 / 0.5 pts

1. Is the following definition correct?

$\Omega(g) = \{f : \exists c \in \mathbb{R} \text{ and } \exists n_0 \geq 0 \text{ that } \forall n \geq n_0 : c \cdot g(n) \leq f(n) \text{ holds } \}$

[Select]

2. Select which logical operation fits in place of the question mark to make the statement true!

$f_1 \in O(g_1) \wedge f_2 \in O(g_2)\ ?\ f_1 * f_2 \in O(g_1 * g_2)$

a) $\rightarrow$ b) $\leftarrow$ c) both d) neither

[Select]

3. Select the appropriate asymptotic notation! What fits in place of the question mark?

$(\varepsilon > 0 \text{ constant})\ n^{\varepsilon} \in\ ?(lg\ n)$? = [Select]

$(c, d > 1 \text{ constant})\ d^{n} \in\ ?(n^{c})$? = [Select]

4. The following hypothetical running times are given. List them in increasing order according to their asymptotic complexity! Let us denote "asymptotically less" with "~<".

(a) $(n^2 + 1)logn$

(b) $5n^2(n + \sqrt{n})$

(c) $2\sqrt{n^2} + n \log n$

(d) $3n(n + 3)$

(c) ~< [Select] ~< [Select]

~< [Select]

Given a real matrix of n x m items (n is the number of rows, m the number of columns), we want to use the programming theorems we have learned to test whether the maximum values of the rows give a monoton increasing series. What can we say about the complexity of mT(n,m) and MT(n,m)? Choose the correct one.

mT(*n*,*m*) in [Select]

MT(*n*,*m*) in [Select]

Answer 1:

Incorrect

Answer 2:

(a)

Answer 3:

Omega

Answer 4:

Omega

Answer 5:

(c)

Answer 6:

(d)

Answer 7:

(a)

Answer 8:

(b)

Answer 9:

Theta(m)

Answer 10:

Theta(n*m)

Partial

Question 2

0.35 / 0.5 pts

1. Is the following definition correct?

$\Omega(g) = \{f : \exists c > 0 \text{ and } \exists n_0 \geq 0 \text{ that } \forall n \geq n_0 : c \cdot f(n) \leq g(n) \text{ holds } \}$

[Select]

2. Select which logical operation fits in place of the question mark to make the statement true!

$f \in \Theta(g)$? $f \in O(g) \wedge f \in \Omega(g)$

a) $\rightarrow$ b) $\leftarrow$ c) both d) neither

[Select]

3. Select the appropriate asymptotic notation! What fits in place of the question mark?

$(c, d > 1 \text{ constant})\ n^c \in\ ?(d^n)$? = [Select]

$(c, \varepsilon > 0 \text{ constant})\ n^{c+\varepsilon} \in\ ?(n^c * lg\ n)$? =

[Select]

4. The following hypothetical running times are given. List them in increasing order according to their asymptotic complexity! Let us denote "asymptotically less" with "~<".

(a) $2n + log(n^2)$

(b) $3n * \log(2^n)$

(c) $\lg n * (2n + \lg n)$

(d) $(n + 2) * \sqrt{n}$

[Select] ~< [Select] ~<

[Select] ~< [Select]

Given an n x m real matrix. The minimum values selected from the rows are sorted using insertion sort. What can we say about the asymptotic complexity of mT(n,m) and MT(n,m)? Select the correct option!

mT(*n*,*m*) in [Select]

MT(*n*,*m*) in Theta(n*(n+m))

Answer 1:

Not correct

Answer 2:

(c)

Answer 3:

O

Answer 4:

Omega

Answer 5:

(a)

Answer 6:

(b)

Answer 7:

(c)

Answer 8:

(d)

Answer 9:

Theta(n*m+n)

Answer 10:

Theta(n*(n+m))

Quiz Score: **0.85** out of 1